Credit Risk: Part III

Lecture 7_1

BANK3011

DR WEI CUI

THE UNIVERSITY OF SYDNEY
Business School



Objectives of this week's lecture

- To examine how credit risk can be managed using:

- Traditional Credit Management Techniques
  - ✓ As Background to examine the traditional market for the sale of loans as background
    - ➢ How it operates in the US
    - ➢ The motivation

Objectives of this week's lecture

- Advanced Credit Management Techniques
  - ✓ Asset Securitisation
    - ➢ Structures
    - ➢ Significance for bank management
  - ✓ An introduction to Credit Derivatives
    - ➢ What are CD's
    - ➢ Use of CD's in credit management

1. Managing credit risk

- Credit risk measurement is required at both the individual loan level as well as at the portfolio level
  - ✓ At the individual loan level the bank will monitor the creditworthiness of individual loans, generally for large loans only.
  - ✓ At the portfolio level the bank will monitor the concentration risk within the portfolio.
  - ✓ This is one of the key economic functions we noted in Week 1 that banks perform.

1. Managing credit risk

- To a large extent however the ability of banks to manage this risk has been limited
  - ✓ Traditional way is to transfer loans to someone else
  - ✓ The loan sales market has been significant in the US

1. Managing credit risk

- Types of Loan Sales Contracts
  - ✓ Loan sales have taken place for over 100 years
  - ✓ Historically this is the simplest way to transfer credit risk
  - ✓ Participations
    - ➢ Loan Syndications are an important example of participations
      - ▪ A participation in a syndicated loan is often transferrable
    - ➢ There is however limited contractual control in these structures

1. Managing credit risk

- Types of Loan Sales Contracts
  - ✓ Assignments (the US Loan Sales Market)
    - ➢ The bulk of the US market (90% +)
    - ➢ All rights transferred on sale of loan
    - ➢ Complexity e.g. the treatment of accrued interest
    - ➢ Loans are essentially not readily marketable or saleable assets

- Participants in the US Loan Sales Market
  - ✓ Buyers
    - ➢ Buyers of distressed loans are generally investment banks, hedge funds or vulture funds.
    - ➢ Inter-bank loan sales in the traditional market were historically due to US branching restrictions.
    - ➢ Foreign banks are dominant buyers of domestic USD loans
    - ➢ Insurance companies and pension funds seeking long-term investments.
    - ➢ Mutual or managed funds and even non-financial firms

1. Managing credit risk

- Participants in the US Loan Sales Market
  - ✓ Sellers
    - ➢ Major money center banks, Foreign banks, U.S. government and its agencies (Freddie Mac etc.), ‘Good Bank-Bad Bank’ models

1. Managing credit risk

- The motivations for loan sales
  - ✓ To meet the reserve requirement
  - ✓ To boost reported income
  - ✓ To meet the capital requirement
  - ✓ Liquidity risk

1. Managing credit risk

- Factors deterring future loan sales growth
  - ✓ Access to commercial paper market
  - ✓ Customer relationship effects
    - ➢ Customers may take a negative view on having their loans sold to another party
  - ✓ Informational concerns – though credit ratings on loans offered for sale have helped to reduce this
  - ✓ Legal concerns - Fraud risk etc.

2. Advanced Methods to Manage Credit Risk

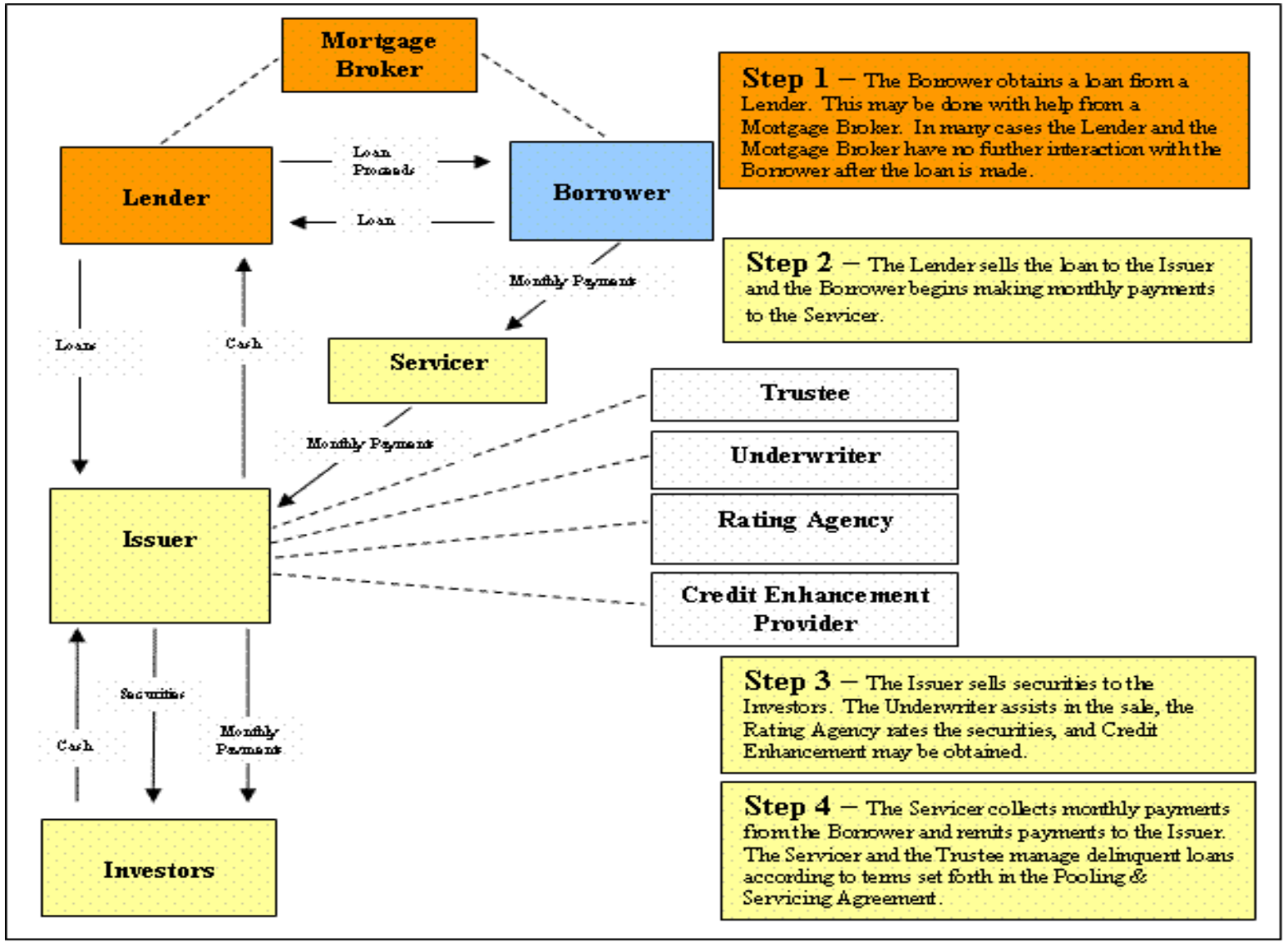
- Asset Securitisation & Credit Derivatives
  - ✓ Both represent different techniques for removing all or part of the credit exposures from a bank's balance sheet
    - ➢ Thus enabling credit risk to be managed by altering the portfolio's concentration or the extent of diversification
    - ➢ Both represent advances over traditional loan sales because they create marketable instruments
      - ▪ Either cash or derivative instruments
  - ✓ They can also be used to add specific types of credit risk to a portfolio thus enhancing portfolio diversification for FIs

- Other Motives and Consequences of a Bank Undertaking Loan Securitisation
  - ✓ Duration management
    - ➢ The duration gap is affected since assets (and their duration) are removed from the balance sheet of the bank sponsoring the securitisation
  - ✓ Liquidity management
    - ➢ Liquidity is affected because cash is received by the bank initiating the Securitisation
  - ✓ Capital management
    - ➢ Capital requirements are reduced as risk-weighted assets decline

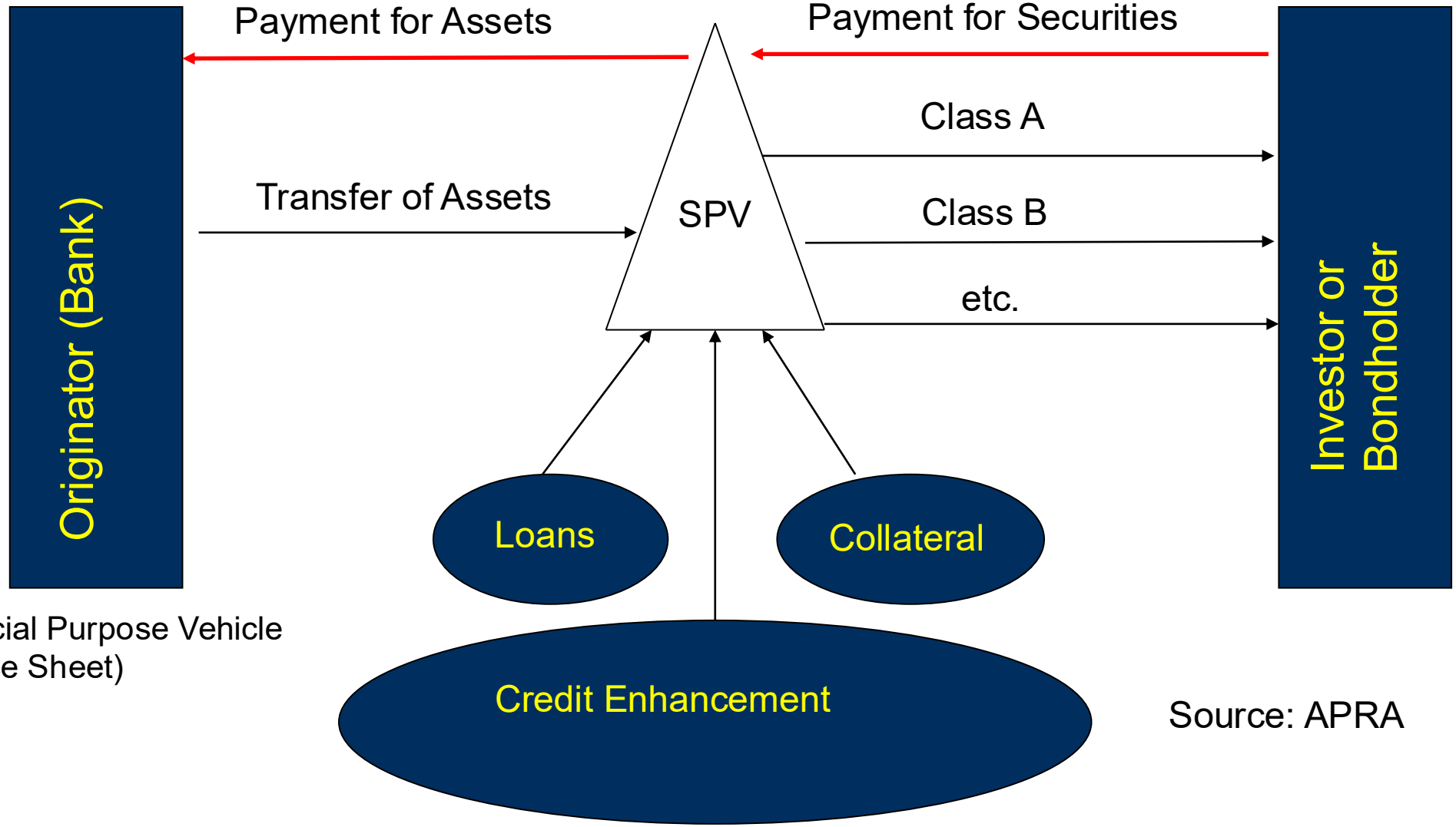
- Other Motives and Consequences of a Bank Undertaking Loan Securitisation
  - ✓ Funding
    - ➢ The bank replaces the funding and its cost via Securitisation
  - ✓ Generates Fee Income for the FI
    - ➢ Fees are received for the ongoing management of the Securitisation
    - ➢ Any profits from the sale are recorded as trading income

- Types of Cashflow Securitisation
  - ✓ Pass-Through Securities (Ginnie Mae, Freddie Mac)
  - ✓ Mortgage or Asset Backed Bonds
  - ✓ Collateralised Debt/Mortgage Obligation

- The Mechanism of Pass-Through Security



- Three Broad Types of Cashflow Securitisation



SPV= Special Purpose Vehicle
(Off Balance Sheet)

Source: APRA

- The Special Purpose Vehicle (SPV)
  - ✓ The role of the SPV is to manage the assets (loans and securities) owned by the SPV and is the issuer of the CDO securities to the investors
  - ✓ The SPV is usually a trust, which means that whilst it legally owns the assets, it holds them for the benefit of the investor or bondholder (the investor is the beneficial owner)

- The Special Purpose Vehicle (SPV)
  - ✓ The SPV must be separate from the bank. The bank may act as manager of the SPV (if so earning a fee for performing such services) but cannot own or control the SPV. This means that the SPV's assets no longer belong to or are controlled by the Bank.
    - ➢ Thus the bank does not need to consolidate this entity for accounting purposes and the assets are no longer assets of the bank – THEY HAVE BEEN SOLD

- CMO or CDO - Classes
  - ✓ The classes in a CMO can be established on the basis of:
    - ➢ Repayment risk
      - e.g. Class A are repaid first when any underlying loans are repaid early, Class B are not repaid until all Class A are and so on
    - ➢ Default risk
      - e.g. Class C incur any credit losses first, then in turn the other classes once class C has been exhausted (Class C is subordinated and is in effect quasi-equity)
    - ➢ A combination of both

- Asset Securitisation - Credit Enhancement
  - ✓ The Credit quality of the securitised assets can be enhanced through one or a combination of the following:
    - ➢ Lenders Mortgage insurance
      - Generally applies where the Loan to Value Ratio (LVR) exceeds 80%
      - Two principal providers in Australia – Genworth and QBE
    - ➢ Guarantees or Letters of Credit
      - Issued by the originating financial institution or another FI e.g. insurance company

- Asset Securitisation - Credit Enhancement
  - ✓ The Credit quality of the securitised assets can be enhanced through one or a combination of the following:
    - ➢ Credit derivatives (see the following slides)
      - ▪ Credit derivatives in essence provide a guarantee because the protection seller is obligated to compensate the buyer for any losses

2. Advanced Methods to Manage Credit Risk

- Typical Credit Derivatives
  - ✓ Total Rate of Return Swaps
  - ✓ Credit Default Swaps
  - ✓ Other types of credit derivatives
    - ➢ Basket Swaps
    - ➢ Credit Spread Options

2. Advanced Methods to Manage Credit Risk

- The Australian Credit Derivative Market Turnover (AUD million)

Transaction	2004-05	2008-09	2009-10	2011-12	2012-13	2013-14	2014-15
Credit Default Swaps	45,330	185,685	108,610	121,031	48,153	65,136	67,570
Total Rate of Return Swaps	0	143	320	228	389	0	-
Basket Swaps and Credit Indices	1,867	146,223	137,873	289,669	180,066	164,551	57,863
Total	46,179	332,051	246,803	410,927	228,608	229,687	125,433

Source: AFMA

3. Total Rate of Return Swap (TROR)

- An agreement in which the total return of an underlying credit sensitive asset or pool of assets is exchanged for some other cashflow

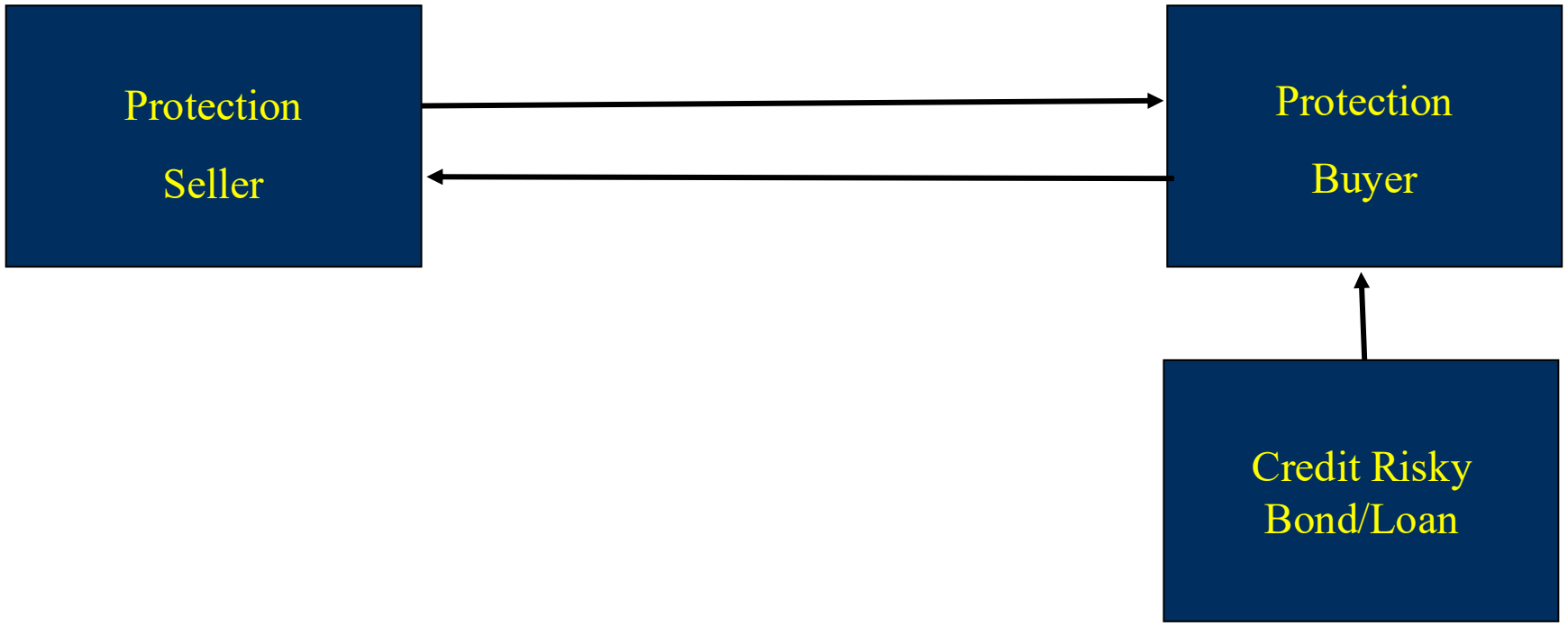
- No principal is exchanged & no change in physical ownership occurs

- Contract specifies notional amount and the dates on which payments will be made

Source: AFMA

3. Total Rate of Return Swap (TROR)

- TROR Swap in Practice
  - ✓ Protection buyer pays fixed rate+ Total Return on the credit risky bond/loan
  - ✓ Protection seller pays Interest (LIBOR) + spread + losses on the credit risky bond/loan



3. Total Rate of Return Swap (TROR)

- Effects of a TROR Swap
  - ✓ The protection buyer effectively goes short of the asset (sells it) without losing ownership
  - ✓ The protection seller effectively goes long of the asset (buys it)
  - ✓ The seller assumes the economic risks of the credit risky asset, but not the asset itself, i.e. the asset is not acquired
- Attributes of a TROR Swap
  - ✓ The maturity need not match that of the underlying
  - ✓ Can include caps and/or floors on the rate
  - ✓ They can be used to exploit an arbitrage opportunity

4. Credit Default Swap

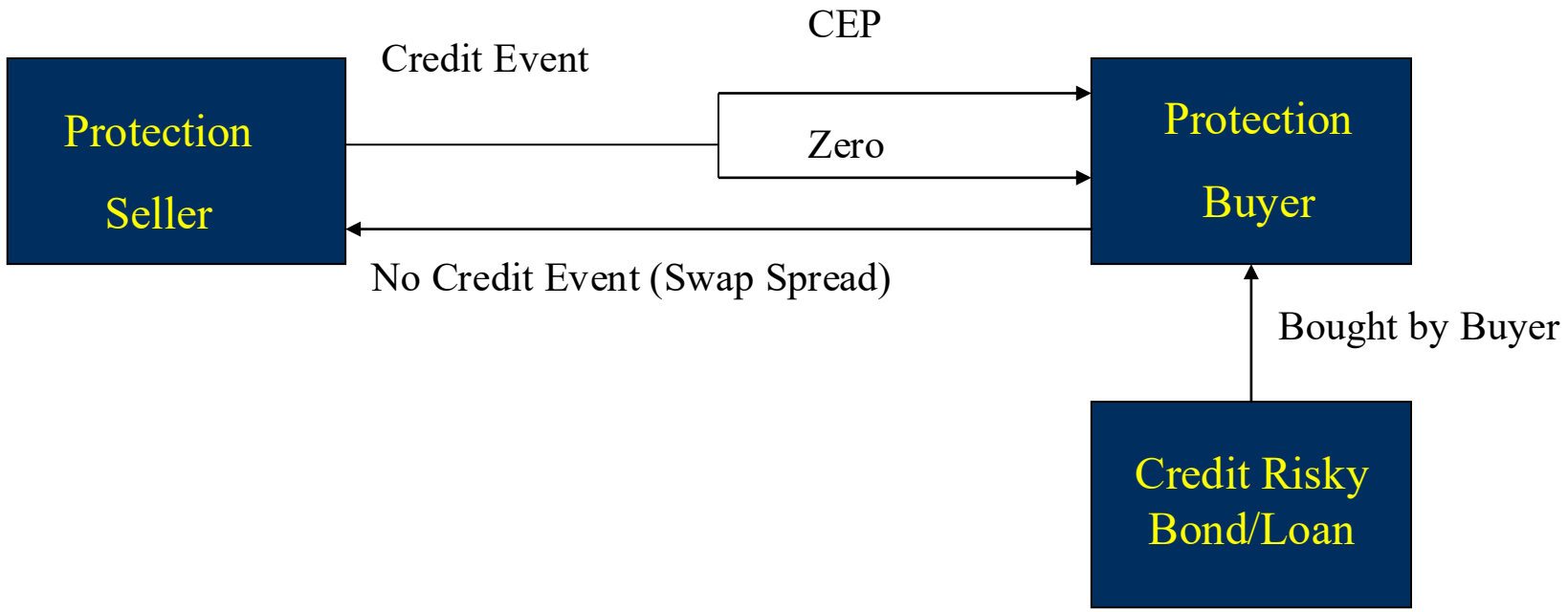
- An agreement in which a periodic or upfront payment is made by the protection buyer in return for the promise to be paid out (i.e. insured) if a predetermined credit event occurs

- No principal is exchanged & no change in physical ownership occurs

- Contract specifies notional amount and the credit events which will trigger payments

4. Credit Default Swap

- Credit Default Swap in Practice
  - ✓ Protection seller makes the credit event payment (CEP) if a specified credit event occurs
  - ✓ Protection buyer pays the agreed basis points x notional principal (the swap spread)



4. Credit Default Swap

- Attributes of Credit Default Swaps
  - ✓ Ideally suited to be able to manage portfolio concentration risk
  - ✓ Credit default swaps can be either on single name credits or on a “basket” or index of cashflows
  - ✓ Terms are generally for one year and longer though the 5 year tenor is usually the most liquid
  - ✓ Credit risk is effectively transferred between the parties

4. Credit Default Swap

- Using CDS' for Credit Risk Management

✓ Buying protection reduces portfolio concentration or individual loan exposure

➢ By effectively removing the asset's credit risk from the portfolio

✓ Sell credit protection to diversify risk

➢ By effectively adding the asset's credit risk to the portfolio

➢ No funding required

✓ **Note CDS creates counterparty risk exposure**

5. Summary

- Summary of Managing Credit Exposure
  - ✓ Traditionally credit risk exposure is reduced by selling the loan/bond to remove it from the balance sheet
  - ✓ Modern/Alternative Risk Reduction Strategies
    - ➢ Originating a CDO is a more effective way of selling loans
    - ➢ Alternatively can reduce exposure to an obligor or group of obligors by buying protection using:
      - TROR Swaps or a Credit Default Swap (CDS)
      - Basket Swaps (bundle of credits)

Sovereign Risk

Lecture 7_2

BANK3011

DR WEI CUI

THE UNIVERSITY OF SYDNEY
Business School



Objectives of this week's lecture

- This session discusses FIs' exposure to sovereign risk
  - ✓ Compare sovereign risk with credit risk
  - ✓ Discuss the typical forms of sovereign default
  - ✓ Introduce various models of country-risk evaluation

1. Introduction

- Early 2000s:
  - ✓ Recovery from the 1997-98 Asian and Russian crises
  - ✓ Turkey and Argentina experienced financial crises

- Late 2000s – GFC
  - ✓ Developed countries faced some of the worst declines in GDP ever experienced
  - ✓ GIIPS sovereign debt crisis – starting from Ireland then Greece and spreading to Portugal, Spain, and Italy

1. Introduction

- Late 2000s – GFC
  - ✓ IMF pledged to inject $250 billion
  - ✓ Downgrade of 9 European countries' sovereign ratings by S&P in January 2012
  - ✓ Downgrade of the U.S. sovereign rating by S&P in August 2011

3. Credit Risk vs. Sovereign Risk

- Sovereign default: Governments can impose restrictions on debt repayments to outside creditors
  - ✓ Maybe independent of the borrower's credit quality.
  - ✓ Legal remedies are very limited.

- Sovereign risk is dependent on both:
  - ✓ The willingness and the ability of the borrower to repay the loan,
  - ✓ The probability that the Government of the country in which a foreign borrower operates will allow the repayment of foreign debt.

3. Credit Risk vs. Sovereign Risk

- Foreign Banks Vs. Hungary
  - ✓ Following market opening (connected to EU access), foreign banks financed Hungarian firms and citizens (including over USD 15 billion in mortgage-related loans between 1998 and 2008) – denominated in euro or Swiss franc.
  - ✓ Hungarian forint devalued (depreciated against Swiss francs by over 40% between 2008 and 2010).

3. Credit Risk vs. Sovereign Risk

- Foreign Banks Vs. Hungary
  - ✓ Hungarian government passed a law to allow mortgage borrowers to repay loans at an exchange rates about 25% below the current market rates in 2011.
  - ✓ Accused of expropriation (loss around USD 1.7 billion).

3. Credit Risk vs. Sovereign Risk

- Foreign Banks Vs. Croatia
  - ✓ In 2015, newly elected Croatian government passed a law converting approximately 3.4 billion of outstanding Swiss franc loans into Euros
  - ✓ UniCredit Bank Austria AG and Zagrebačka Banka DD (two subsidiaries of UniCredit SPA) accused Croatia's government of expropriation (loss around $1.2 billion)
  - ✓ The two foreign lenders filed a complaint at the **ICSID** alleging violation of the 1997 **BIT between Croatia and Austria**

3. Credit Risk vs. Sovereign Risk

- A two-step decision in terms of credit risk analysis
  - ✓ Assess the credit quality of the borrower
  - ✓ Assess the sovereign risk quality of the country in which the borrower resides

4. Sovereign ceiling effect

- Typically, there is a sovereign ceiling effect in that no borrower from a country can be better rated than the sovereign obligor of that country

- Exceptions may be multinational firms operating in multiple countries with parents head-quartered in countries with much lower sovereign risk than the host countries for its subsidiaries

5. Debt Rescheduling v.s. Debt Repudiation

- Debt repudiation: outright cancellation of all current and future debt obligations by a borrower
  - ✓ Since WWII, only China, Cuba, and North Korea have repudiated debt
  - ✓ Recent steps to forgive debts of most severe cases conditional on reforms targeted to improve poverty problems, which avoids debt repudiation

- Debt rescheduling: changing the contractual terms of a loan, such as its maturity and interest payments
  - ✓ Most common form of sovereign risk events
  - ✓ South Korea 1998; Argentina 2001; Ongoing EU debt crisis – Greece

5. Debt Rescheduling v.s. Debt Repudiation

- After World War II, most international debtor problems were in the form of debt rescheduling

- Contributing factors
  - ✓ Most debts are in the form of bank loans rather than bonds – fewer lenders involved
  - ✓ Often loan syndicates comprised of same group of FIs – cohesive lenders
  - ✓ Cross-default provisions – prevent selective defaults on weak lenders

7. Country Risk Evaluation - Outside evaluation models

- The Euromoney Index
  - ✓ Now an index based on a large number of economic and political factors (scores: 0=max risk, 100 = no risk)
- The Economist Intelligence Unit ratings
  - ✓ A combined economic and political risk rating on a 100-point scale (max risk = 100)
- Institutional Investor Index
  - ✓ Based on the scores given by the loan officers of major multinational banks surveyed
  - ✓ Scale: 0 = certainty of default, 100 = no default possibility

8. Country Risk Evaluation (Internal) - Statistical models

- Credit scoring models based on primarily economic ratios
- The selected variables are tested for predictive power in separating rescheduling countries from non-rescheduling countries using past data.
- Commonly used economic ratios and relation with the probability of default (like debt-rescheduling):
  - ✓ (positive) Debt service ratio: (Interest + amortization on debt)/Exports
  - ✓ (positive) Import ratio: Total imports / Total FX reserves
  - ✓ (likely negative) Investment ratio: Real investment / GNP
  - ✓ (positive) Variance of export revenue = σ^2_{ER}
  - ✓ (positive) Domestic money supply growth = $\Delta M/M$

9. Problems with Statistical Country Risk Assessment Models

- Measurements of key variables
  - ✓ Lack of up-to-date information and inaccurate prediction
- Population groups
  - ✓ Finer distinction beyond reschedulers and non-reschedulers may be necessary
- Political risk factors
  - ✓ Not captured in most CRA models
  - ✓ Strikes, corruption, elections, revolution
  - ✓ Corruption Perceptions Index

9. Problems with Statistical CRA Models

- Portfolio aspects
  - ✓ Traditional country-by-country CRA models ignore the diversification effect.
    - ➢ Systematic default risks across countries
  - ✓ Differentiate between the systematic component and country-specific component of factors contributing to defaults

- Incentive aspects
  - ✓ Most CRA models lack a sophisticated analysis of the benefits versus costs of defaults for both borrowers and lenders.

10. Country Risk Evaluation (Internal) Using Market Data

- The secondary market for LDC (less-developed-countries) debts
  - ✓ Quoted prices provide information about investors' perception about default risk in these debts.
  - ✓ Sellers:
    - ➢ Large FIs willing to accept write-downs of loans to reduce their risk exposure
    - ➢ Small FIs wishing to disengage from LDC loan market,
    - ➢ FIs willing to swap LDC debt to rearrange portfolios.

10. Country Risk Evaluation (Internal) Using Market Data

- Buyers:
  - ✓ Wealthy investors, hedge funds/vulture funds, FIs seeking to engage in debt-for-equity swaps or speculative investments;
  - ✓ FIs seeking to rebalance their foreign debt portfolios.

- Market segments
  - ✓ Sovereign bonds
  - ✓ Brady bonds
  - ✓ Performing LDC loans
  - ✓ Nonperforming LDC loans

10. Country Risk Evaluation (Internal) Using Market Data

- Repayment problems are likely to be predicted by combining LDC debt prices with key variables.

- Examples of significant variables affecting LDC loan sale prices (P)
  - ✓ Debt service ratios
  - ✓ Import ratio
  - ✓ Accumulated debt in arrears
  - ✓ Amount of loan loss provisions against LDC loans.

- Approach subject to same criticisms as traditional statistical models.

11. Mechanisms for Dealing with Sovereign Risk Exposure

(1) Loan sales

- Benefits:
  - ✓ Removal from balance sheet, free up resources
  - ✓ Signaling effect: the rest of the balance sheet is strong enough to absorb cost

- Cost: the loss itself

(2) Loan-for-bond swaps

- Transform LDC loan into marketable liquid instrument (bond)
- Benefits:
  - ✓ Bonds: marketable, low transaction cost, small bid-ask spread, efficient clearing and settlement system
  - ✓ For Brady bonds, have full or partial collateral backing ->Usually senior to remaining loans of that country
- Costs:
  - ✓ Bonds usually have longer time to maturity
  - ✓ Loan swap less than 1 dollar for 1 dollar

12. Mechanisms for Dealing with Sovereign Risk Exposure: MYRAs

(3) Debt-for-equity swaps

(4) Multiyear restructuring agreements (MYRAs) – restructure the original debts.

Benefits and costs depend on:

- ✓ Fee charged by FIs for restructuring
- ✓ Interest rate charged on the new loan
- ✓ Grace period for payments on new loan
- ✓ Maturity of new loan – normally lengthened
- ✓ Option and guarantee features

- Concessionality: the amount a bank gives up in PV terms as a result of a MYRA.

Summary

- We have discussed the exposure to sovereign risk of FIs.
  - ✓ Compare sovereign risk with credit risk
  - ✓ Discuss the typical forms of sovereign default
  - ✓ Introduce various models of country-risk evaluation